

What's New in Displays for Embedded Systems

Various challenges, such as physical size and power consumption, have become more of a concern for embedded devices. This article presents some high-level solutions that are available to tackle the challenge

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We have all seen the transition from CCFT- to LED-backlit LCD modules, and the many benefits it provides for design engineers working on incorporating display technology into their devices. Modern displays, such as TFT and OLED, are evolving with promising designs for tomorrow, including the integration of new high-speed interfaces to handle the emergence of very-high-resolution imaging needs and the possible elimination of backlight in E-Ink displays. Overall, displays for embedded systems have evolved well enough to help design efficient and better products, often directly affecting the duration of the design cycle itself. Let's see how.

Ready-to-deploy solutions

The choice of which embedded display to use could be a complex decision to figure out, especially for those who are new to integrating an embedded display into their design. Vendors have announced new all-in-one display products that cater to the requirements of a design engineer developing from industrial to consumer applications, with the main aim being to make the design process faster. These displays are engineered to integrate anything from power supplies to integrated processors. They include all the necessary components and integrate various functions like switch, UART, USB, SPI, I²C and PWM, along with SD card storage, to allow for large storage capabilities.

Ravi Pagar, regional director-South Asia, element14 India, says, "The all-in-one offerings reduce time-to-market and development costs. Engineers can quickly integrate multiple functionalities into their next design and also upgrade their designs, quite simply. These ready-to-deploy solutions can be adopted for products in various areas like industrial, consumer electronics and telecommunications."



VIA launches new series of rugged LED backlit displays for embedded applications

Upgrade for mono STN LCD modules. If you do not have much experience on TFT module development, and you want to upgrade your mono LCD display to a TFT, there is no need to worry about microcontroller unit (MCU) limitations. TFTs with integrated processors are available for upgrading mono displays to colour TFT displays, saving R&D engineers time and accelerate the product development procedures.

These latest 'clever' TFTs do not require any modification to the exist-

ing firmware and circuit on your PCB. For instance, Winstar's WF35M allows the designer to implement colour TFT without having to redesign the TFT firmware and circuit. You can easily switch to TFT with a simple procedure, avoiding too many I/O, without using an expensive high-performance CPU or MCU. These TFTs eliminate the need for using FPC at the client's end, thereby adding to the savings for your bill of material (BOM) costs for final products.

Onboard video processing for easier integration.

Systems having a combination of video processing functionality built directly into the screens have made it possible for easier integration with a number of applications. These display modules, like 4D System's 4DLCD-FT843 and FTDI's VM800B, are highly advanced intelligent SPI displays, based on integrated video engine and amplifier-enabled audio, providing low-cost display solutions for your embedded designs.

The integrated engine offloads the host processor and provides a variety of graphics features. Besides, various interfacing options allow different hosts to be connected directly to the displays, providing a powerful set of audio and graphics features to the host using the on-board audio/video engine. These modules are best suited for requirements of various HMI applications, including industrial control terminals, intelligent instruments, data acquisition and analysis, medical products and network terminals.